

Project Olympus · Unit 1: The Language of Change

Teacher's Edition with answer keys, hints, and solutions · The Language of Everything · David Balyeat, with Claude

Three weeks. A function is a machine. A graph is a picture of a relationship. Slope is a rate. An equation is a question, and solving it is the answer. By the Labor you'll know whether Lamine Yamal can catch Cristiano Ronaldo with a head start, and you'll prove it three ways: algebra, a graph, and a spreadsheet.

Week 1 · Day 1: The Machine

Functions: something goes in, a rule happens, something comes out

Warm-up

A server earns a **\$16.50/hour** base. Four hours of work: base pay?

Answer: \$66. 16.5×4 . You just ran a function: hours in, dollars out.

The Story

A function is a machine

A vending machine: you press B4, you get the same chips every time. Press B4 and get a different snack each time and you'd call it broken. That reliability is the whole idea of a **function**: something goes in (the input), a rule happens, and exactly one thing comes out (the output). Same input, same output, every time.

DECODER

Function

A rule that turns each input into exactly one output. Input $\rightarrow$ rule $\rightarrow$ output. If you know the rule and the input, you know the output.

"Multiply by 3 and add 2" is a function. Put in 4, get 14. Put in 4 again tomorrow, still 14.

DECODER

Expression

A math phrase with no equals sign, like $3x + 2$. It's a recipe, not a statement. You can evaluate it (plug in a number) but you can't "solve" it, because it isn't claiming anything.

$3x + 2$ is an expression. $3x + 2 = 17$ is an equation (it claims something, and it can be true or false).

Function notation: $f(x)$

Mathematicians name their machines. **$f(x) = 3x + 2$** means: "the machine called f , when you feed it x , gives back $3x + 2$." Then **$f(4)$** means "feed it 4," and $f(4) = 3(4) + 2 = 14$.

Read $f(x)$ as "f of x." **It is not f times x.** This confuses everyone for about a week and then never again. The parentheses here mean "input goes here," like a slot.

$$f(x) = 3x + 2 \quad f(4) = 3(4) + 2 = 14 \quad f(10) = 32$$

WORKED EXAMPLE · THE SERVER PAY MACHINE

A 5-hour shift at \$16.50/hour is \$82.50 base. Tips are about 18% of what the tables spend. So pay depends on sales, s :

$$P(s) = 82.5 + 0.18s$$

If the tables spend \$600: $P(600) = 82.5 + 0.18(600) = 82.5 + 108 = \mathbf{\$190.50}$. If they spend \$0 (dead night): $P(0) = \$82.50$. The machine takes sales in and hands pay out. (Base rate is a stand-in; tipped wages vary by state. The 18% is real; that's what people tip.)

WORKED EXAMPLE · THE STORM MACHINE

In Fortnite's first storm phase the storm does **1 HP per second**. Your health if you stand in it: $H(t) = 100 - 1 \cdot t$, where t is seconds. $H(30) = 70$. $H(100) = 0$, which is the game's polite way of saying you should have rotated.

By phase 4 it's 7 HP/s: $H(t) = 100 - 7t$. Same machine, meaner rule.

THE STORY · WHO NAMED IT

Gottfried Leibniz used the word "function" in 1673 (he also invented calculus, which we'll get to, and had a legendary feud with Newton about it). Leonhard Euler gave us the $f(x)$ notation in 1734. Euler wrote so much math that publishing his collected work took over a century. He also went blind and just kept going, dictating it.

MATHEMAGIC · THE TRICK IS A CHAIN OF MACHINES

"Double it, add 10, halve it, subtract the original" is four machines in a row: $d(n) = 2n$, then $a(n) = n + 10$, then $h(n) = n \div 2$, then $s(n) = n - (\text{original})$. Feeding the output of one machine into the next is called **composing** them. Every card trick you'll learn this year is a composition: shuffle, cut, deal, and somewhere in the chain the machines cancel out so the card ends up where you knew it would.

ON PAPER

Codex: *function*, *expression*, $f(x)$. Then draw three machines from your life as boxes with an arrow in and an arrow out: e.g. "jijitsu classes per week $\rightarrow$ hours on the mat per

month."

Practice

1. $f(x) = 3x + 2$. Find **$f(5)$** .

Answer: 17

Hints: (1) Replace every x with 5. (2) $3(5) + 2$.

Solution: $3 \times 5 + 2 = \mathbf{17}$.

2. $g(x) = x^2 - 1$. Find **$g(4)$** .

Answer: 15

Hints: (1) Replace x with 4: $4^2 - 1$. (2) $4^2 = 16$.

Solution: $16 - 1 = \mathbf{15}$.

3. Server pay: $P(s) = 82.5 + 0.18s$. A slow night, \$400 in sales. **$P(400) = ?$**

Answer: 154.5 dollars (± 0.01)

Hints: (1) 0.18×400 first. (2) 72. Then add the base.

Solution: $82.5 + 72 = \mathbf{\$154.50}$.

4. Fortnite phase 3 storm: 5 HP/s. $H(t) = 100 - 5t$. Your health after 12 seconds in the storm: **$H(12) = ?$**

Answer: 40 HP

Hints: (1) $5 \times 12 = 60$.

Solution: $100 - 60 = \mathbf{40 \text{ HP}}$. Rotate.

5. Spider-Man's web cartridge holds **200 shots**. Each swing uses **3**. Which rule gives shots left after n swings?

A. $S(n) = 200 + 3n$

B. $S(n) = 200 - 3n$

C. $S(n) = 3n - 200$

D. $S(n) = 200 \div 3n$

Answer: $S(n) = 200 - 3n$ (B)

Hints: (1) Shots go *down* as swings go up. (2) Start at 200, lose 3 per swing.

Solution: Start with 200, subtract 3 for each swing: **$S(n) = 200 - 3n$** .

6. Using $S(n) = 200 - 3n$, how many shots are left after **50 swings**?

Answer: 50

Hints: (1) 3×50 .

Solution: $200 - 150 = \mathbf{50 \text{ shots}}$. Time to reload. (Tom Holland's Peter would have run out sooner; he wastes a lot on quips.)

7. In Minecraft, walking speed is **4.317 blocks per second** (a block is one meter). Distance walked: $D(t) = 4.317t$. **$D(60) = ?$** (Nearest whole block.)

Answer: 259 blocks (± 1)

Hints: (1) 4.317×60 . (2) Estimate first: $4 \times 60 = 240$, and a bit more.

Solution: $4.317 \times 60 = \mathbf{259 \text{ blocks}}$ in a minute of walking.

8. In Harry Potter's world, 29 Knuts make a Sickles and 17 Sickles make a Galleon, so a Galleon is $17 \times 29 = 493$ Knuts. Knuts in g Galleons: $K(g) = 493g$. **$K(3) = ?$**

Answer: 1,479 Knuts

Hints: (1) 493×3 . (2) Estimate: $500 \times 3 = 1,500$, minus a bit.

Solution: $493 \times 3 = \mathbf{1,479 \text{ Knuts}}$. Goblin accountants have a hard job.

Apply: Machines in a Sheet

Sheets is a function machine with a memory. Build one.

1. New tab in your Olympus sheet. A1: "x". B1: " $f(x) = 3x + 2$ ".
2. A2 through A11: the numbers 1 to 10. (Type 1 and 2, select both, drag the corner down.)
3. B2: $=3*A2+2$. Press Enter. Drag B2's corner down to B11. You just evaluated $f(1)$ through $f(10)$ in two seconds.
4. Column C: the server pay machine. C1: "P(s)". Put sales 0, 100, 200 ... 1000 in column D, and in C: $=82.5+0.18*D2$, filled down.
5. Column E: the storm. E2: $=100-5*A2$. Where does it hit zero? (It doesn't in 10 seconds. Extend A to 20.)

Dad notes: Make sure the formulas reference cells (A2), not typed numbers. Ask him: "If I change the 3 to a 5, what happens to the whole column?" and have him do it.

Week 1 · Day 2: Pictures of Relationships

Descartes, the fly on the ceiling, and why a graph is worth a thousand numbers

Warm-up

In your head: the point **(3, -2)** is where, relative to the axes?

- A. Right of the y-axis, above the x-axis
- B. Right of the y-axis, below the x-axis
- C. Left of the y-axis, below the x-axis
- D. Left of the y-axis, above the x-axis

Answer: Right and below. $x = 3$ means 3 steps right; $y = -2$ means 2 steps down.

The Story

THE STORY · THE FLY ON THE CEILING

René Descartes liked to sleep late. The story (possibly true, definitely told a lot) is that he was lying in bed one morning in 1637 watching a fly wander around the ceiling, and it hit him: he could describe exactly where the fly was with just two numbers, its distance from one wall and its distance from another. Two numbers, one point. And if the fly moved in a pattern, the pattern could be written as an equation.

That's the **Cartesian plane** (named after him). It's the bridge between algebra (equations) and geometry (pictures). Before Descartes they were separate subjects. After him, every equation had a shape and every shape had an equation.

DECODER

Coordinate plane

Two number lines crossing at right angles. The horizontal one is the x-axis, the vertical one is the y-axis, and the crossing point is the origin (0, 0). Every point gets an address: (x, y), always x first.

(3, 25.5) means "3 across, 25.5 up." On a distance-time graph that could be "at 3 seconds, 25.5 meters."

A graph is a picture of a function

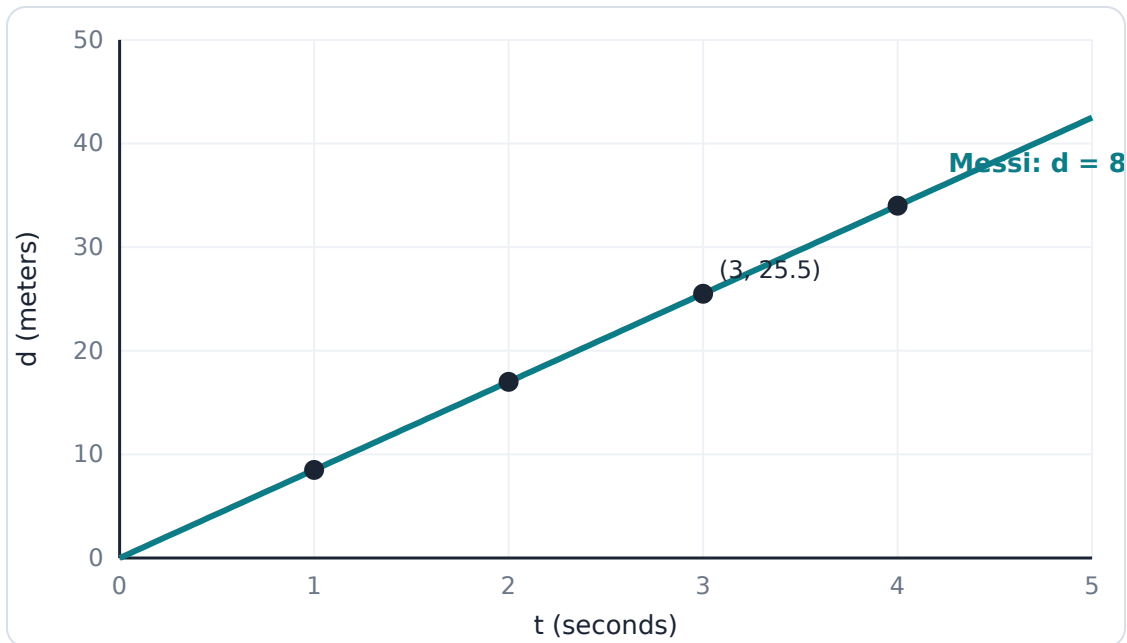
Take a machine, feed it lots of inputs, and plot every (input, output) pair as a point. Connect them. That picture *is* the function. You can read outputs off it without calculating, see where it's rising or falling, and compare two machines at a glance.

WORKED EXAMPLE · MESSI'S SPRINT

Messi's top recorded speed at the 2026 World Cup was 30.9 km/h, about 8.5 m/s. Suppose he's already at top speed and sprinting for the ball. Every second, 8.5 more meters.

t (seconds)	0	1	2	3	4
d (meters)	0	8.5	17	25.5	34

The machine is $d(t) = 8.5t$. Plot it:



Straight line. That's what constant speed looks like. Read it: at $t = 2$, $d = 17$. Where is he at $t = 3.5$? Halfway between 25.5 and 34: about 30. You didn't calculate, you *looked*.

DECODER

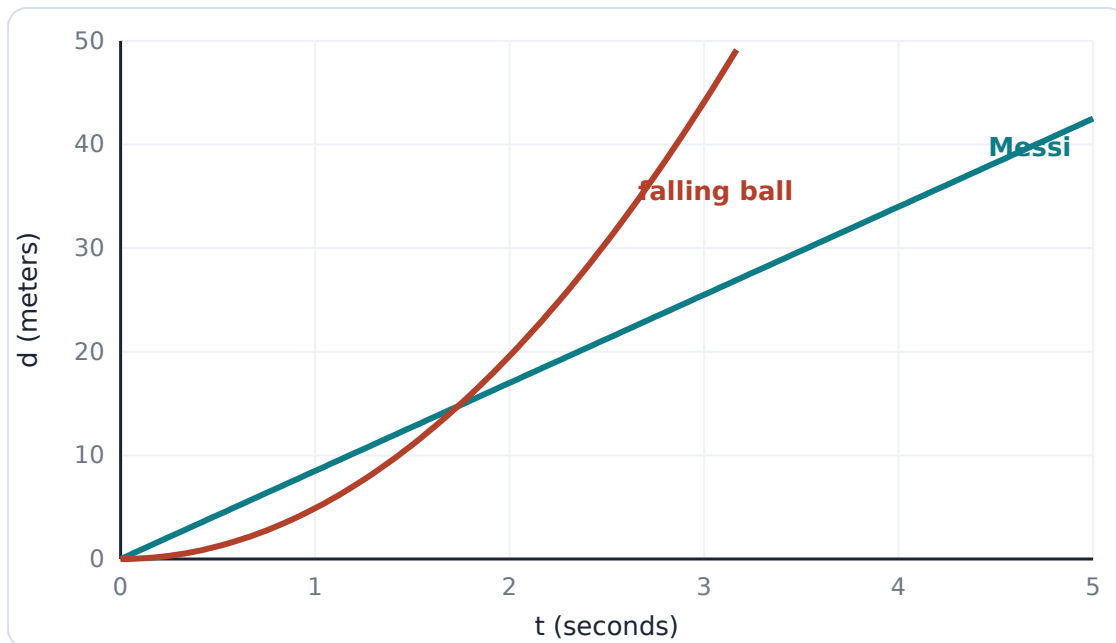
Linear

A function whose graph is a straight line. It changes by the same amount every step. Constant speed is linear. A fixed hourly wage is linear. Storm damage in one phase is linear.

$d = 8.5t$ adds 8.5 every second. Linear. $d = 4.9t^2$ adds more each second (a falling ball). Not linear.

WORKED EXAMPLE · NOT EVERYTHING IS A LINE

Drop a ball off a Fortnite cliff (or jump from the Battle Bus with your glider closed). Distance fallen is $d = 4.9t^2$. Compare it to Messi:



The ball starts slow and gets faster; the line bends. That bend is **acceleration**, and it's Unit 4. For now: straight line means steady rate. Curve means the rate is changing.

GREEK ORIGINS · WHY IS IT CALLED GEOMETRY?

Geo = earth, *metron* = measure. The Greeks (and the Egyptians before them) invented it to re-survey farmland after the Nile flooded every year and washed away the boundary markers. Descartes' trick was realizing that measuring the earth and doing algebra were the same activity. Thales, Euclid, and Archimedes had the shapes; Descartes gave them coordinates.

ON PAPER

Graph paper. Plot Messi's table by hand: axes labeled with units, points marked, line drawn with a ruler. Then plot Lamine Yamal (35.5 km/h $\approx$ 9.9 m/s) on the same axes: at $t = 1, 2, 3, 4$ he's at 9.9, 19.8, 29.7, 39.6. Which line is steeper?

Practice

1. In the point **(4, 9)**, what is the x-coordinate?

Answer: 4

Hints: (1) x always comes first.

Solution: 4. (Alphabetical: x before y.)

2. $f(x) = 2x + 1$. Which point is on its graph?

- A. (3, 7)
- B. (3, 6)
- C. (7, 3)
- D. (2, 4)

Answer: (3, 7) (A)

Hints: (1) Plug the x-value in and see if you get the y-value. (2) $f(3) = 2(3) + 1 = ?$

Solution: $f(3) = 7$, so **(3, 7)** is on the graph. (7, 3) is not: $f(7) = 15$.

3. Messi at 8.5 m/s: how far at $t = 4$?

Answer: 34 m

Hints: (1) $d = 8.5t$.

Solution: $8.5 \times 4 = \mathbf{34 \text{ m}}$. Also readable straight off the table.

4. From the graph of $d = 8.5t$: at what time is Messi at **25.5 m**?

Answer: 3 s

Hints: (1) Find 25.5 on the vertical axis, go across to the line, drop down. (2) Or: $25.5 \div 8.5$.

Solution: **3 seconds**. Reading a graph backwards (output $\rightarrow$ input) is something you'll do constantly.

5. Yamal at **9.86 m/s** (35.5 km/h, fastest of the Spain squad at the 2026 World Cup): how far after 3 seconds? One decimal.

Answer: 29.6 m (± 0.1)

Hints: (1) 9.86×3 .

Solution: **29.6 m**. About 4 meters ahead of Messi at the same moment. Messi would say he doesn't need to sprint; the ball comes to him.

6. Two distance-time lines on the same axes: Yamal's and Messi's. Which is steeper, and why?

- A. Yamal's, because he covers more meters per second
- B. Messi's, because he has more goals
- C. They're the same; both are straight
- D. Neither; steepness depends on the units

Answer: Yamal's, because he covers more meters per second (A)

Hints: (1) Steeper = more rise per step across.

Solution: **Yamal's**. More meters per second means the line climbs faster. Steepness is speed. Tomorrow has a name for that.

7. Every point that sits *on the y-axis* has what x-coordinate?

Answer: 0

Hints: (1) The y-axis is the vertical line through the origin. How far left or right is it?

Solution: 0. Points on the y-axis look like (0, something). The place a line crosses the y-axis will matter a lot in Session 5.

Apply: Desmos: your first graph

Open **desmos.com/calculator** (free, no account needed). It's a graphing calculator that lives in a browser.

1. Type $y = 8.5x$. There's Messi.
2. Below it type $y = 9.86x$. There's Yamal.
3. Type $(3, 25.5)$. A point appears. Is it on Messi's line?
4. Click the wrench (top right) to set the axes: x from 0 to 5, y from 0 to 50.
5. Now type $y = 4.9x^2$. Watch the ball fall. Where does it cross Messi's line? (Hover the intersection; Desmos will show you.)
6. Screenshot, print or sketch it into the notebook next to your hand-drawn version.

Dad notes: The hand-drawn one matters more than the Desmos one. Check that the axes have units and the scale is even (each grid square worth the same amount). Uneven scales are the number one way kids lie to themselves with graphs.

Week 1 · Day 3: Slope Is a Rate

Rise over run, and why it is secretly velocity

Warm-up

Convert **36 km/h** to meters per second.

Answer: 10 m/s. 36 km = 36,000 m; an hour = 3,600 s; $36,000 \div 3,600 = 10$. Shortcut you'll use forever: **km/h $\div$ 3.6 = m/s.**

The Story

Slope: how steep, and in which direction

Yesterday you saw that Yamal's line is steeper than Messi's because he covers more meters per second. Today we measure steepness.

DECODER

Slope

How much a line rises for each step it runs. Slope = rise $\div$ run = (change in y) $\div$ (change in x). Pick any two points on the line and compute it; you'll get the same number.

From (1, 8.5) to (3, 25.5): rise = $25.5 - 8.5 = 17$, run = $3 - 1 = 2$, slope = $17 \div 2 = 8.5$.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}}$$

DECODER

Δ (delta)

The Greek letter D, used to mean "change in." Δy is "the change in y." It's not a variable; it's a label that says "subtract the before from the after."

If your health goes from 100 to 40, $\Delta H = 40 - 100 = -60$. Negative, because it went down.

DECODER

Rate

A change in one thing per unit of another. Meters per second. Dollars per hour. HP per second. Every rate is a slope, and every slope is a rate.

8.5 m/s is a rate. On a distance-time graph it's the slope. Same thing, two names.

Slope is velocity

Here's the payoff. On a distance-vs-time graph, rise is meters and run is seconds, so slope is meters per second. **The slope of a position graph is the velocity.** Not "related to." Is. This

is the first sentence of physics and the first sentence of calculus, and you just learned it from Messi.

In *Project Hail Mary*, Ryland Grace spends half the book staring at graphs and reading slopes off them to figure out how fast Astrophage is growing or how fast the Hail Mary is going. The whole plot runs on this.

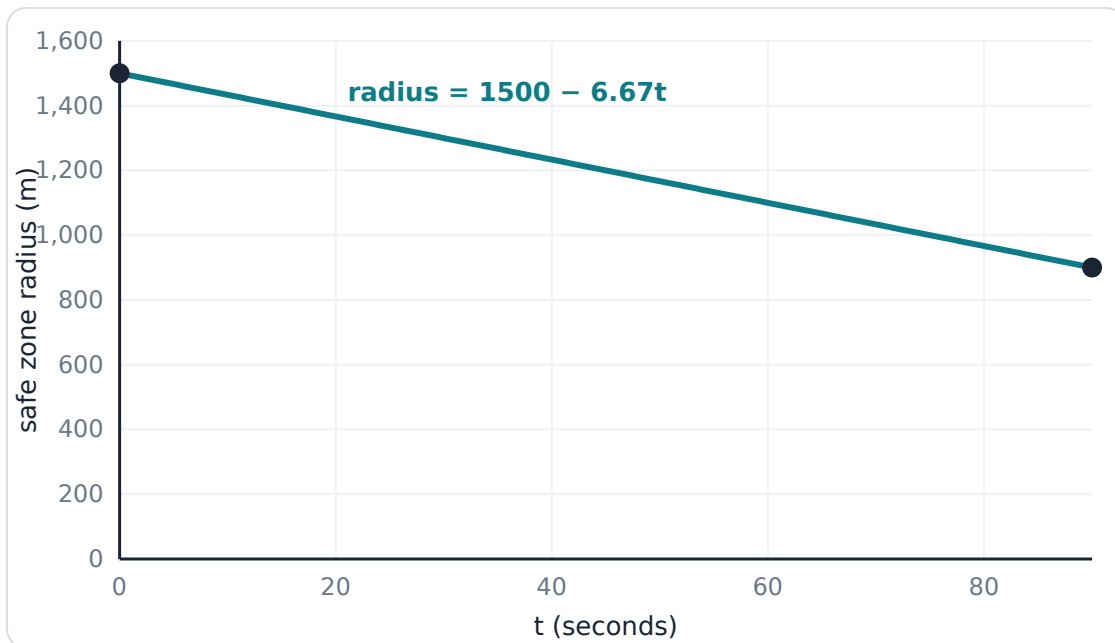
WORKED EXAMPLE • THE 2026 WORLD CUP SPEED BOARD (REAL DATA, FIFA/SOFASCORE)

Player	km/h	÷ 3.6 → m/s
Kylian Mbappé	37.6	10.44
Achraf Hakimi	35.6	9.89
Erling Haaland	35.6	9.89
Lamine Yamal	35.5	9.86
Harry Kane	31.4	8.72
Lionel Messi	30.9	8.58

Cristiano Ronaldo's famous number is 33.95 km/h, clocked at the 2018 World Cup when he was 33 years old. He wasn't on the 2026 board. He is 41.

WORKED EXAMPLE • NEGATIVE SLOPE: THE STORM

Fortnite's first storm circle takes 90 seconds to shrink (that's the real Chapter 7 timing). Say the safe zone's radius goes from 1,500 m down to 900 m in those 90 seconds. Slope = $(900 - 1,500) \div 90 = -600 \div 90 = \mathbf{-6.67 \text{ m/s}}$. Negative, because the radius is falling. The storm edge moves inward at about 6.7 meters per second. You sprint at maybe 6 m/s. Do the math on whether to loot that chest.



MAT MATH · THE SHOT

A double-leg takedown: you drop your level and drive about 1.5 meters into your partner in about half a second. Average speed = $1.5 \div 0.5 = \mathbf{3 \text{ m/s}}$. Slow compared to Messi, but Messi isn't trying to move a person. Thursday's Lab Day: time five of your own shots (phone stopwatch, or better, slow-mo video) and get your real number. That's the Mat Scientist badge.

ON PAPER

Codex: *slope*, Δ , *rate*. Draw a rise-over-run triangle on yesterday's Messi graph, between $t = 1$ and $t = 3$. Label rise and run with units.

Practice

1. Slope of the line through **(1, 8.5)** and **(3, 25.5)**.

Answer: 8.5 (± 0.01)

Hints: (1) Rise: $25.5 - 8.5$. Run: $3 - 1$. (2) $17 \div 2$.

Solution: $17 \div 2 = \mathbf{8.5}$. Messi's speed, read off two points.

2. Storm phase 3: health goes from **(0 s, 100 HP)** to **(20 s, 0 HP)**. Slope?

Answer: -5 HP/s (± 0.01)

Hints: (1) Rise = $0 - 100$ (it's negative). (2) Run = 20.

Solution: $-100 \div 20 = \mathbf{-5 \text{ HP/s}}$. Matches the phase 3 damage rate. Slope of health vs. time is damage rate.

3. Convert Ronaldo's **33.95 km/h** to m/s. Two decimals.

Answer: 9.43 m/s (± 0.02)

Hints: (1) Divide by 3.6.

Solution: $33.95 \div 3.6 = \mathbf{9.43 \text{ m/s}}$. Hold onto this; he's in the Labor.

4. Storm phase 2 shrinks for 75 seconds. Say the radius goes from **1,200 m to 600 m**. Slope, in m/s?

Answer: -8 m/s (± 0.01)

Hints: (1) $(600 - 1200) \div 75$.

Solution: $-600 \div 75 = \mathbf{-8 \text{ m/s}}$. Faster than you can run. Leave early.

5. A double-leg shot: **1.5 m in 0.5 s**. Average speed?

Answer: 3 m/s (± 0.01)

Hints: (1) Distance $\div$ time.

Solution: **3 m/s**. (Slope of your position graph during the shot.)

6. Minecraft sprint-jumping averages **7.127 m/s**. The line through (0, 0) and (10, ?): what's the missing y? Two decimals.

Answer: 71.27 blocks (± 0.05)

Hints: (1) Slope $\times$ run = rise. (2) 7.127×10 .

Solution: **71.27 blocks** in 10 seconds. Sprint-jumping is also the most efficient way to travel per hunger point, which is a slope of a different graph.

7. Line A has slope 4. Line B has slope 9. Both are distance vs. time. Which is faster?

A. A

B. B

C. Same speed

D. Can't tell without the intercept

Answer: B (B)

Hints: (1) Slope is speed.

Solution: **B**. 9 meters per second beats 4. The intercept (where it starts) doesn't change the speed.

8. A distance-time line with **zero slope** (flat) means:

- A. Standing still
- B. Moving backward
- C. Infinite speed
- D. Moving at 1 m/s

Answer: Standing still (A)

Hints: (1) No rise means no change in distance.

Solution: Standing still. Distance isn't changing as time passes. A vertical line would be "infinite speed," which is why you'll never see one on a real position graph.

Apply: The speed board in Sheets

Turn the World Cup table into a working model.

1. New tab: "Speed board." Column A: the six players plus Ronaldo. Column B: km/h (real numbers from the story).
2. C2: $=B2/3.6$. Fill down. That's m/s.
3. D2: $=30/C2$. Fill down. That's seconds to sprint 30 meters at top speed. Format to 2 decimals.
4. Select A1:D8, Data → Sort range by column D. Who's first? Who's last?
5. Insert → Chart. Make a bar chart of column C. Title it. Screenshot it into the notebook.
6. Desmos: type $y = mx$, click "add slider" for m. Drag m from 0 to 12. Watch the line tilt. That slider is speed.

Dad notes: Ask: "If I add a player at 40 km/h, what happens?" (He should just add a row. If he wants to retype formulas, the lesson hasn't landed.) Also: Bolt's 100 m record averaged about 37.6 km/h over the whole race, with a peak near 44. Mbappé's 37.6 is a peak, not an average. Good dinner-table fact.

Week 1 · Day 4 · Lab Day: Lab Day: Galileo's Ramp

Measure motion with your own hands, plot it, and find the slope

Warm-up

A ball rolls **2 meters in 4 seconds**. Average speed?

Answer: 0.5 m/s. Distance ÷ time. "Average" because it may have been speeding up the whole way, which is exactly what you're about to see.

The Story

THE STORY · GALILEO SLOWS DOWN GRAVITY

Around 1600, Galileo wanted to know how falling things move. Problem: they fall too fast to time, and there were no stopwatches. So he tilted the problem: roll a ball down a gentle ramp, which is falling in slow motion. For a clock, he let water flow from a jug into a cup while the ball rolled, then *weighed the water*. More water, more time. Crude, brilliant, and it worked.

He found something strange: the ball didn't cover equal distances in equal times. In the second second it went three times as far as in the first; in the third second, five times. Distance grew with the **square** of time. Nobody had ever measured that. It's the first equation of physics, and it's Unit 4's whole plot. Today you just need to see the curve with your own eyes.

The experiment

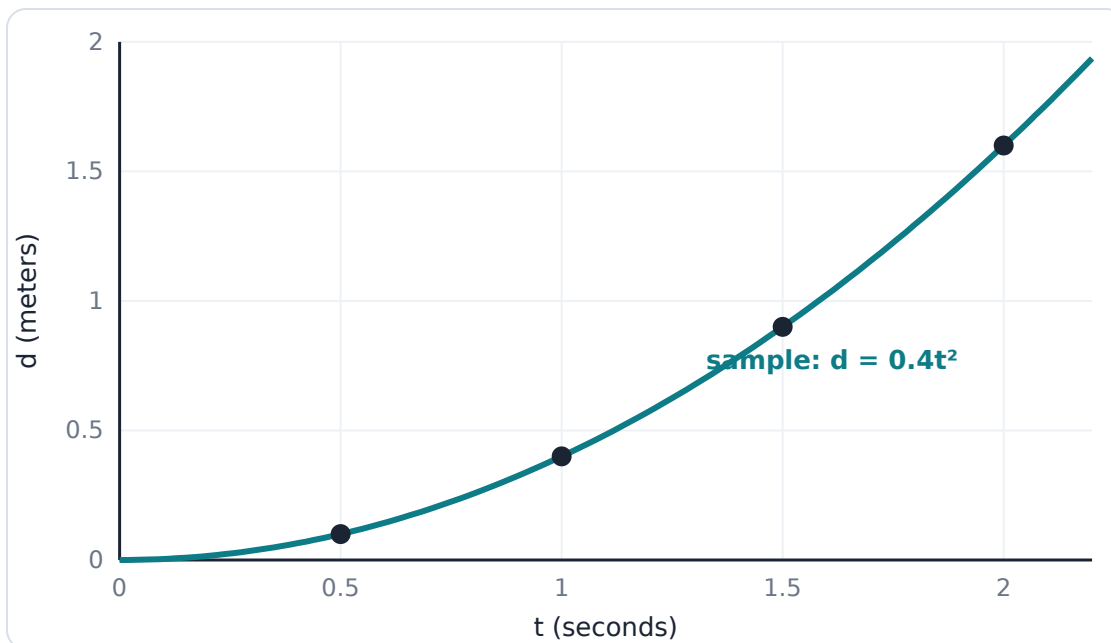
Materials: a board, shelf, or long Lego baseplate as a ramp (1 to 2 meters), propped a few centimeters high at one end (gentle!); a marble, golf ball, or Rubik's-cube-sized ball; masking tape; a tape measure; a phone with a stopwatch, or better, slow-motion video.

- 1 Tape marks every 20 cm along the ramp from the top: 0, 20, 40, ... up to 160 cm or whatever fits.
- 2 Release the ball from the top (don't push it). Start the timer at release.
- 3 Record the time when it passes each mark. Easiest with slow-mo video: film once, then scrub through and read the phone's timestamps. Stopwatch works too; just repeat each run 3 times and average.
- 4 Table in your notebook: t (s) and d (m). Convert cm to m.
- 5 Plot d vs. t on graph paper. Points, then connect them. Does it curve?
- 6 Compute the slope between the *last two* points and between the *first two*. Different? That difference is Galileo's discovery.

WORKED EXAMPLE · WHAT YOU SHOULD SEE (SAMPLE DATA)

t (s)	0	0.5	1.0	1.5	2.0
d (m)	0	0.1	0.4	0.9	1.6

Your numbers will differ (steeper ramp, faster), but the *shape* should match: bending upward. Use this sample data for the Practice problems if your run isn't finished yet.



Sheets: let the computer find the slope

Type your t values in column A and d values in column B. Then in any cell: `=SLOPE(B2:B6, A2:A6)`. Sheets fits the best straight line through your points and gives its slope: your *average* speed over the run. (Note the order: y-range first, then x-range. Everyone gets this backwards once.) Then insert a scatter chart and look at the curve.

MAT MATH · MAT MATH DATA NIGHT

At jiu-jitsu this week: film five of your shots or a sprint across the mat in slow-mo. Mark the start and end points; measure the distance with a tape (or count mat squares; a standard mat tile is 1 m). Read the time off the video. Compute your speed for each. Enter them in the sheet; `=AVERAGE()` them. That's real data, and it's the Mat Scientist badge.

Practice

1. Using the sample data: average speed over the whole run (1.6 m in 2.0 s)?

Answer: 0.8 m/s (± 0.01)

Hints: (1) Distance $\div$ time.

Solution: $1.6 \div 2 = \mathbf{0.8 \text{ m/s}}$.

2. Slope between the **last two** sample points, (1.5, 0.9) and (2.0, 1.6)?

Answer: 1.4 m/s (± 0.01)

Hints: (1) Rise: $1.6 - 0.9$. Run: 0.5.

Solution: $0.7 \div 0.5 = \mathbf{1.4 \text{ m/s}}$.

3. Slope between the **first two** sample points, (0, 0) and (0.5, 0.1)?

Answer: 0.2 m/s (± 0.01)

Hints: (1) $0.1 \div 0.5$.

Solution: **0.2 m/s**. Seven times slower than at the end. Same ball, same ramp.

4. The slope keeps increasing along the run. The ball is:

- A. Speeding up
- B. Slowing down
- C. Moving at constant speed
- D. Stopped

Answer: Speeding up (A)

Hints: (1) Slope is speed. Increasing slope means...

Solution: **Speeding up**. That's acceleration, and it's why the graph curves instead of drawing a line. Galileo saw exactly this.

5. Times are in A2:A6 and distances in B2:B6. Which formula gives the slope?

- A. `=SLOPE(A2:A6, B2:B6)`
- B. `=SLOPE(B2:B6, A2:A6)`
- C. `=SLOPE(A2:B6)`
- D. `=RISE(B2:B6)/RUN(A2:A6)`

Answer: `=SLOPE(B2:B6, A2:A6)` (B)

Hints: (1) Sheets wants the y-values first.

Solution: **`=SLOPE(B2:B6, A2:A6)`**: y-range, then x-range. Distance is y (it depends on time), time is x.

Apply: Run it, plot it, sheet it

The Practice above used sample data. Now do it for real.

1. Run the ramp experiment. At least 5 marks, at least 2 runs each.

2. Hand-plot in the notebook. Ruler for axes. No ruler for the curve.
3. Enter the data in a new "Ramp" tab. `=SLOPE()` for the average speed. Insert a scatter chart.
4. Write two sentences in the log: what the curve means, and one thing that went wrong in the measurement.
5. **Beat Dad Friday:** go to the Beat Dad page. Week 1's problem is live. Dad logs in as himself for his turn.
6. Check the cloud chip at the top says **saved**. (No cloud? Save a progress file to the Math Program folder.)

Dad notes: Keep the ramp gentle (5 to 10 cm rise over a meter) or the times are too short to measure. Slow-mo video on the iPhone is 240 fps, which is plenty. If the curve isn't obvious, the ramp is too steep or the marks are too close together. Mat Scientist badge: award it once he brings real timing data from class into the sheet.

Week 2 • Day 1: $y = mx + b$

Every straight line has a starting point and a rate. Learn to read both.

Warm-up

$P = 82.5 + 0.18s$. If sales are **zero**, what's the pay?

Answer: \$82.50. The base. That number has a name today: the intercept.

The Story

The anatomy of a line

$y = mx + b$ m = slope (the rate) b = y-intercept (where it starts)

DECODER

y-intercept

Where the line crosses the y-axis: the value of y when $x = 0$. In a real situation it's the starting amount, the flat fee, the head start, the base pay.

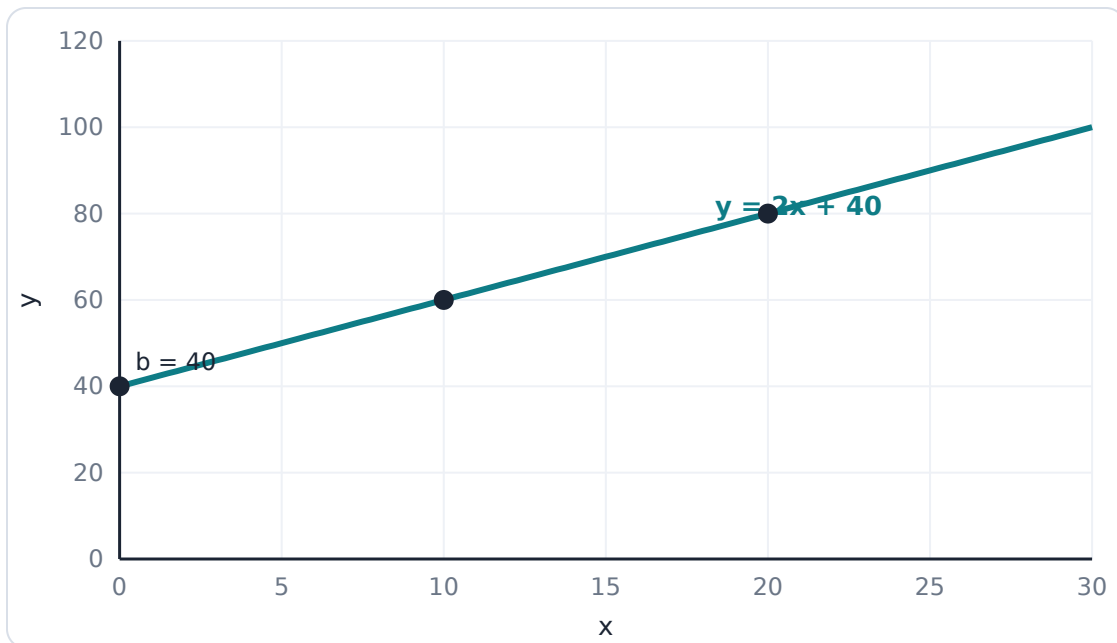
In $P = 0.18s + 82.5$, the intercept is 82.5: what you're paid before a single table tips. In $r = 1500 - 6.67t$, it's 1500: the radius when the shrink starts.

Every linear situation in the world has these two parts. **b** answers "where does it start?" and **m** answers "how fast does it change?" Once you can spot them in a sentence, you can write the equation in ten seconds.

Situation	b (start)	m (rate)	Equation
Server: \$82.50 base + 18% of sales	82.5	0.18 per dollar	$P = 0.18s + 82.5$
Storm: 1,500 m radius, shrinking 6.67 m/s	1500	-6.67 per second	$r = -6.67t + 1500$
Filament: \$20 per kg plus \$5 shipping	5	20 per kg	$C = 20k + 5$
Rubik's: 60 s average, improving 2 s per week	60	-2 per week	$T = -2w + 60$

WORKED EXAMPLE • READING IT OFF A GRAPH

Given a line, find b by looking where it hits the y-axis. Find m with a rise-over-run triangle anywhere on it. Below: the line crosses at 40 and climbs 20 for every 10 across, so $m = 2$ and the equation is $y = 2x + 40$.



GREEK ORIGINS · ZEUS REFINANCES OLYMPUS

Zeus takes out a loan to renovate the throne room (gold leaf, a bigger lightning-bolt rack). He owes 12,000 drachmas and pays 400 a month. What he still owes after n months: $D = -400n + 12,000$. Intercept: the original debt. Slope: negative, because it's shrinking. He'll be debt-free at $n = 30$. Hera thinks the rack was unnecessary.

Real loans have interest, which makes this not a line. That's Unit 2, and it's why the bank always wins.

ON PAPER

Codex: y -intercept, slope-intercept form. Then write three $y = mx + b$ situations from your own life (phone plan? allowance plus chores? belt promotions?) with b and m labeled.

Practice

1. $y = 4x + 7$. The slope is:

Answer: 4

Hints: (1) The number multiplying x .

Solution: 4.

2. $y = 4x + 7$. The y -intercept is:

Answer: 7

Hints: (1) The number by itself: y when $x = 0$.

Solution: 7.

3. A line passes through **(0, 12)** with slope **3**. Its equation?

- A. $y = 12x + 3$
- B. $y = 3x + 12$
- C. $y = 3x - 12$
- D. $y = x + 15$

Answer: $y = 3x + 12$ (B)

Hints: (1) (0, 12) is on the y-axis, so 12 is the intercept.

Solution: $y = 3x + 12$.

4. A 6-hour shift at \$16.50/hour gives \$99 base, plus 18% of sales. Pay as a function of sales s :

- A. $P = 99s + 0.18$
- B. $P = 0.18s + 99$
- C. $P = 18s + 99$
- D. $P = 0.18(s + 99)$

Answer: $P = 0.18s + 99$ (B)

Hints: (1) Start (b) is the base. Rate (m) is the tip fraction. (2) $18\% = 0.18$.

Solution: $P = 0.18s + 99$. Watch out for 18 vs 0.18: an 18 would mean tips of \$18 per dollar spent, which is a restaurant you should open.

5. Filament: $C = 20k + 5$ (dollars, k in kg). Cost of **3.5 kg**?

Answer: 75 dollars (± 0.01)

Hints: (1) 20×3.5 , then add 5.

Solution: $70 + 5 = \text{\$75}$. About enough filament for 35 Cartographer badge tokens, or one very large Lego brick.

6. Rubik's average: $T = 60 - 2w$. After **10 weeks**?

Answer: 40 seconds

Hints: (1) $2 \times 10 = 20$.

Solution: $60 - 20 = \textbf{40 s}$. (The world record is around 3 seconds. The line can't go on forever; at some point it has to bend. Everything linear is linear only for a while.)

7. Storm: $r = 1500 - 6.67t$. Radius after **60 seconds**? Nearest meter.

Answer: 1,100 m (± 1)

Hints: (1) 6.67×60 . (2) About 400.

Solution: $1500 - 400 = \textbf{1,100 m}$ (1,099.8 to be fussy).

8. Which line is **decreasing** (goes down as x increases)?

- A. $y = 2x + 5$
- B. $y = -2x + 5$
- C. $y = 5$
- D. $y = x$

Answer: $y = -2x + 5$ (B)

Hints: (1) Negative slope.

Solution: $y = -2x + 5$. ($y = 5$ is flat: slope 0.)

Apply: Sliders

Desmos makes m and b into physical objects you can push around. Do that until it's boring.

1. Desmos: type $y = mx + b$. Click "add all sliders."
2. Drag b . The line moves up and down without tilting. That's the start.
3. Drag m . The line tilts around the intercept. That's the rate. Make m negative. Make it 0.
4. Now type your three real-life lines from the notebook. Do they look right? Does the intercept make sense?
5. Sheets: a "Line builder" tab. B1 = m , B2 = b . Column A: x from 0 to 10. Column C:
 $=\$B\$1*A2+\$B\2 (the \$ signs lock the cell; you'll learn why by dragging without them).
Chart it. Change B1 and watch the chart tilt.

Dad notes: The $\$B\1 trick (absolute references) is the most useful Sheets skill after fill-down. If he asks why, have him try the formula without the \$ and drag it: it'll grab the wrong cells and the error explains itself.

Week 2 · Day 2: Solving for x

An equation is a balance. Undo what was done. And the trick, explained twice.

Warm-up

$3x = 21$. What is x ?

Answer: 7. Undo the "times 3" by dividing both sides by 3. That's the whole method, just with more steps.

The Story

The balance

An equation says two things weigh the same. $3x + 2 = 17$ says "three x 's and a 2 balance a 17." To find x , you're allowed to do *anything* to the scale, as long as you do it to both sides. Take 2 off both sides: $3x = 15$. Split both sides into thirds: $x = 5$. The scale never tipped, so the answer is honest.

DECODER

Solve

Find the value of the variable that makes the equation true. Every legal move keeps both sides equal; the goal is to get the variable alone on one side.

$3x + 2 = 17 \rightarrow 3x = 15 \rightarrow x = 5$. Check: $3(5) + 2 = 17$. ✓ Always check. It takes five seconds and catches everything.

DECODER

Inverse operation

The move that undoes another move. Subtracting undoes adding. Dividing undoes multiplying. Solving is running the machine backwards: undo the last thing first.

The machine did $\times 3$ then $+2$. To undo: -2 first, then $\div 3$. Last in, first out, like unpacking a bag.

WORKED EXAMPLE · TWO-STEP, THEN VARIABLES ON BOTH SIDES

$$5x + 4 = 2x + 19$$

Get the x 's together: subtract $2x$ from both sides $\rightarrow 3x + 4 = 19$. Subtract 4 $\rightarrow 3x = 15$. Divide by 3 $\rightarrow x = 5$. Check: $5(5) + 4 = 29$ and $2(5) + 19 = 29$. ✓

Real version: Yamal's distance = Ronaldo's distance. That's Session 9, and it's the Labor.

WORKED EXAMPLE · WHEN DOES THE STORM KILL YOU?

Phase 3: $H = 100 - 5t$. "When is health zero?" is the equation $100 - 5t = 0$. Add $5t$ to both sides: $100 = 5t$. Divide: **$t = 20$ seconds**. Every "how long until" question is an equation waiting to be solved.

MATHEMAGIC · WHY THE TRICK WORKS, AND HOW TO BUILD YOUR OWN

"Double, add 10, halve, subtract original" ends at 5 because $(2n + 10) \div 2 - n = n + 5 - n = 5$. The n cancels. Now you can design tricks:

$$n \rightarrow \times a \rightarrow +b \rightarrow \div a \rightarrow -n = b \div a$$

Pick any a and b . Make b a multiple of a so the answer is a whole number. "Multiply by 4, add 12, divide by 4, subtract your number" $\rightarrow 3$. "Multiply by 5, add 35, divide by 5, subtract your number" $\rightarrow 7$. Add theater: "concentrate on your number... I'm sensing a 7."

Fancier: have them add their *age* somewhere and you can extract it, because you know everything except one thing. Willman badge for a trick you designed yourself, performed, and explained.

GREEK ORIGINS · DIOPHANTUS, AND THE WORD "ALGEBRA"

Diophantus of Alexandria (around 250 AD) wrote the first book that solved equations with symbols instead of words. The word *algebra* came later, from the Arabic *al-jabr* ("restoring"), the title of a book by al-Khwarizmi in Baghdad around 820 AD. Al-jabr meant moving a term to the other side of the balance. You'll do it twenty times today.

Diophantus's tombstone was reportedly a puzzle about his own life. It's Problem 8. Solve it and you know how old he was when he died.

ON PAPER

Codex: *solve, inverse operation*. Design your own number trick using the formula, write the n -table proof, and perform it tonight.

Practice

1. $4x - 6 = 18$

Answer: 6

Hints: (1) Undo the -6 first: add 6 to both sides. (2) $4x = 24$.

Solution: $4x = 24 \rightarrow x = 6$. Check: $24 - 6 = 18$. ✓

2. $\frac{x}{5} + 3 = 10$

Answer: 35

Hints: (1) Subtract 3 first. (2) $x/5 = 7$. Undo the $\div 5$.

Solution: $x/5 = 7 \rightarrow x = 35$.

3. $2x + 7 = 4x - 5$

Answer: 6

Hints: (1) Get x's on one side: subtract $2x$ from both sides. (2) $7 = 2x - 5$. Now add 5.

Solution: $12 = 2x \rightarrow x = 6$. Check: $19 = 19$. ✓

4. Storm phase 4 (7 HP/s): $H = 100 - 7t$. Solve for the time when $H = 30$. One decimal.

Answer: 10 s (± 0.05)

Hints: (1) $100 - 7t = 30$. (2) $70 = 7t$.

Solution: $7t = 70 \rightarrow t = 10$ s.

5. Server: $P = 0.18s + 99$. How much in sales to take home **\$200**? Nearest dollar.

Answer: 561 dollars in sales (± 1)

Hints: (1) $0.18s + 99 = 200$. (2) $0.18s = 101$. Divide.

Solution: $s = 101 \div 0.18 \approx \textbf{\$561}$ in sales. A busy night at a mid-priced place.

6. Rubik's: $T = 60 - 2w$. When does the average hit **30 s**?

Answer: 15 weeks

Hints: (1) $60 - 2w = 30$.

Solution: $2w = 30 \rightarrow w = \textbf{15 weeks}$.

7. Design check: "Multiply your number by 4, add 12, divide by 4, subtract your number." Everyone gets:

Answer: 3

Hints: (1) $b \div a$.

Solution: $12 \div 4 = \textbf{3}$. $(4n + 12) \div 4 - n = n + 3 - n = 3$.

8. Diophantus's tombstone (paraphrased): "He was a boy for one-sixth of his life. After one-twelfth more, he grew a beard. After one-seventh more, he married. Five years later a son was born. The son lived half as long as his father. Four years after the son died, Diophantus died." How old was Diophantus?

Answer: 84

Hints: (1) Call his age x . Write every piece: $x/6 + x/12 + x/7 + 5 + x/2 + 4 = x$. (2) Fractions are annoying. Multiply every term by 84 (the smallest number 6, 12, 7, and 2 all divide into). (3) $14x + 7x + 12x + 420 + 42x + 336 = 84x$. Collect.

Solution: $75x + 756 = 84x \rightarrow 756 = 9x \rightarrow x = 84$. Boy until 14, beard at 21, married at 33, son at 38, son died at 80, Diophantus at 84. Slightly grim for a tombstone, but he'd have liked that you solved it.

Apply: Perform and prove

Your own trick, tonight. The math first, the theater second.

1. Pick a and b (make b a multiple of a). Write the full n -table proof in the notebook.
2. Rehearse the patter. Pauses. Eye contact. "Are you sure you're concentrating?"
3. Perform for Dad. Then hand him the n -table and make him explain it back to you. (Yes, reversed. Teaching is how you find out whether you know it.)
4. Claude: paste your trick's steps and ask "Check my algebra. Don't tell me if it works; tell me if my proof has a gap." See what it says. If it makes a mistake, you know what to do.

Dad notes: Willman badge criteria: a trick he designed (not the one from Unit 0), performed with some showmanship, and explained clearly enough that you could repeat it. Award from the Badges page. If he wants a harder version: have the spectator add their shoe size or age at some step, and have him work out how to extract it.

Week 2 · Day 3: Equations in the Wild

Turning sentences into equations, from Minecraft to Gringotts

Warm-up

Knuts per Galleon: $17 \times 29 = ?$

Answer: 493. $17 \times 30 = 510$, minus $17 = 493$. Estimating with a round number and correcting is faster than long multiplication.

The Story

The translation table

Word problems are just sentences that haven't been translated yet. Certain words always mean the same math:

When the sentence says...	It means...
"per," "each," "every"	a rate: the slope m
"starts at," "flat fee," "already has," "head start"	the intercept b
"how many until," "when will," "how long before"	solve for the variable
"total," "altogether"	add the pieces
"left," "remaining"	start minus what's used
"is," "equals," "the same as"	$=$

DECODER

Model

A math description of a real thing, simplified on purpose. A model leaves stuff out (wind, fatigue, the ref) to keep the math doable, and it's good if it's useful, not if it's perfect. Every equation in this course is a model.

$d = 8.5t$ models Messi at top speed. It ignores that he had to accelerate first. Fine for 3 seconds, wrong for 30.

WORKED EXAMPLE · MINECRAFT

"You need 640 cobblestone for the wall. You mine 1.5 blocks per second. How long?"
Translate: $1.5t = 640$. Solve: $t = 640 \div 1.5 \approx 427$ seconds, about **7 minutes**. (Mining speed is made up; the walking speeds in the problems below are the real ones from the game's code.)

WORKED EXAMPLE · SPIDER-MAN

"200 shots per cartridge, 3 per swing. How many swings until only 20 shots are left?"
Translate: $200 - 3n = 20$. Solve: $3n = 180$, **$n = 60$ swings**. (Peter Parker actually did this math; he's a chemistry nerd who built the web fluid. Mr. Stark just made the suit prettier.)

WORKED EXAMPLE • GRINGOTTS

Wizard money: 29 Knuts = 1 Sickle, 17 Sickles = 1 Galleon. A Skiving Snackbox at Weasleys' Wizard Wheezes for 3 Galleons 10 Sickles, in Knuts: $3 \times 493 + 10 \times 29 = 1,479 + 290 = \mathbf{1,769 \text{ Knuts}}$. The goblins invented this system to make sure nobody could do the math in their head. It worked.

MAT MATH • THE MAT

A jujitsu class runs 6 five-minute rounds with a minute of rest between. Total time on the mat: $6 \times 5 + 5 \times 1 = 35$ minutes. Twice a week, that's 70 minutes of rolling. Over a year (say 48 weeks): $70 \times 48 = 3,360$ minutes = **56 hours** of live sparring. Belts are earned by the hour, not the year. Write your own version with your real class schedule.

ON PAPER

Codex: *model*. Then, for every practice problem below, write the equation on paper *before* solving. The translation is the skill; the solving you already have.

Practice

1. Minecraft walking speed is **4.317 m/s**. How long to walk **1,000 blocks** to your friend's base? Nearest second.

Answer: 232 s (± 1)

Hints: (1) $4.317t = 1000$. (2) Divide.

Solution: $1000 \div 4.317 \approx \mathbf{232 \text{ s}}$, just under 4 minutes.

2. Sprinting is **5.612 m/s**. How many seconds does sprinting save on those 1,000 blocks? Nearest second.

Answer: 53 s (± 1)

Hints: (1) Sprint time: $1000 \div 5.612$. (2) About 178 s. Subtract from the walking time.

Solution: $232 - 178 = \mathbf{53 \text{ seconds saved}}$. Sprint-jumping (7.127 m/s) saves another 38.

3. Web cartridge: $S(n) = 200 - 3n$. After how many swings are **50 shots** left?

Answer: 50 swings

Hints: (1) $200 - 3n = 50$. (2) $3n = 150$.

Solution: **$n = 50$** . Convenient. Peter should refill at 50 and never think about it again.

4. Convert **2 Galleons, 5 Sickles, 20 Knuts** entirely into Knuts.

Answer: 1,151 Knuts

Hints: (1) Galleons $\times$ 493, Sickles $\times$ 29, then add the Knuts. (2) $986 + 145 + 20$.

Solution: **1,151 Knuts.**

5. In *Percy Jackson*, the Gray Sisters' taxi charges (let's say) a **1-drachma** flat fee plus **2 drachmas per mile**. A ride costs **15 drachmas**. How many miles?

Answer: 7 miles

Hints: (1) Flat fee is b, per-mile is m: cost = $2x + 1$. (2) $2x + 1 = 15$.

Solution: $2x = 14 \rightarrow$ **7 miles**. They share one eye, so the driving is priced accordingly.

6. Hudson wants **\$1,000** for a resin printer. Each restaurant shift nets about **\$150**. How many shifts? (Whole shifts; you can't work 0.67 of one.)

Answer: 7 shifts

Hints: (1) $150n = 1000$. (2) $n = 6.67$. Round which way?

Solution: 6.67 shifts, so **7 shifts**: after 6 you'd have \$900. Real problems round up when you need "at least." (Also: \$1,050 after 7. Buy filament with the change.)

7. Your Rubik's average is **45 s**, improving **1.5 s per week**. Weeks until you average **30 s**?

Answer: 10 weeks

Hints: (1) $T = 45 - 1.5w$. Set $T = 30$. (2) $1.5w = 15$.

Solution: **10 weeks**. (Then it gets harder. Improvement is never linear for long. Unit 2 has the curve.)

8. In *Zelda*, say Link runs about **6 m/s** on flat ground. A shrine is **300 m** away. Time to get there?

Answer: 50 s

Hints: (1) $6t = 300$.

Solution: **50 seconds**, if he doesn't stop to cook. He always stops to cook.

Apply: Write two of your own

The best way to learn translation is to write the sentences yourself.

1. Write two word problems from a game or sport you know well. Each needs a start (b), a rate (m), and a "how long until" question.
2. Solve them in the notebook. Check by plugging back in.
3. Give them to Dad. He solves them; you grade him. (Be fair. Be a little bit harsh.)
4. Paste them into Claude and ask for the answers. Then check Claude's work against yours. If it's wrong, screenshot it: Bug Hunter.

Dad notes: Solve them for real; don't sandbag. If one is ambiguous ("you didn't say whether the rate was per minute or per second"), tell him: precision in the problem is half the skill. Bug Hunter: it must be a real error with a demonstrable correct answer, not a matter of interpretation.

Week 2 · Day 4 · Lab Day: Lab Day: The Tip Calculator

Two real Sheets models, one chart, and Beat Dad week 2

Warm-up

18% of \$250?

Answer: \$45. 10% is 25, 5% is 12.50, 3% is 7.50; $25 + 12.5 + 7.5 = 45$. Or just 0.18×250 .

The Story

Build 1: the tip and pay calculator

This is a tool you'll actually use if you ever wait tables (and a version of it if you ever run a business). Inputs at the top, outputs below, and a log of shifts.

	A	B
1	Hours worked	5
2	Base rate (\$/hr)	16.50
3	Sales (\$)	600
4	Tip rate	18%
5		
6	Base pay	<code>=B1*B2</code>
7	Tips	<code>=B3*B4</code>
8	Total	<code>=B6+B7</code>
9	Effective \$/hour	<code>=B8/B1</code>

Type 18% with the percent sign and Sheets stores 0.18. Then a second block, a **shift log**: one row per shift (date, hours, sales), with formulas for pay in each row using `$B$2` and `$B$4` so the rate cells stay locked. `=SUM()` at the bottom. That's a paycheck predictor.

Build 2: the sprint sheet

Two columns of time (0 to 5 seconds in steps of 0.5), then a column for Messi (`=8.58*A2`) and one for Yamal (`=9.86*A2`). Insert a line chart with both. Two lines from the same origin, one steeper. Now add a column for Ronaldo with a 5 m head start: `=9.43*A2+5`. Where does Yamal's line cross Ronaldo's? Hover the chart. Next week you'll compute it exactly.

If you filmed your shots this week, today's the day they go in the sheet: distance, time, and =B2/C2 for speed. Five rows, =AVERAGE() . Compare to the 3 m/s guess from Session 3. Faster? Slower? That's science: a guess, a measurement, and a comparison.

Practice

1. With hours = 5 and base rate = 16.50, what does the Base pay cell (=B1*B2) show?

Answer: 82.5 (± 0.01)

Hints: (1) 5×16.5 .

Solution: 82.50.

2. Sales = 700, tip rate = 18%. The Tips cell shows:

Answer: 126 (± 0.01)

Hints: (1) 700×0.18 .

Solution: 126.

3. In the sprint sheet, seconds for Yamal (9.86 m/s) to cover 30 m: =30/9.86 shows (2 decimals):

Answer: 3.04 s (± 0.01)

Hints: (1) Distance $\div$ speed.

Solution: 3.04 s. That's from a flying start; a real 30 m sprint from standing is closer to 4 s.

4. Same for Messi (8.58 m/s), 30 m, 2 decimals:

Answer: 3.5 s (± 0.01)

Hints: (1) $30 \div 8.58$.

Solution: 3.50 s. Half a second behind Yamal over 30 m. Which, in soccer, is a lot.

5. On the sprint chart at $t = 2$ s, whose line is on top?

- A. Messi
- B. Yamal
- C. They're equal
- D. Depends on the axes

Answer: Yamal (B)

Hints: (1) Higher line = more distance at that time.

Solution: Yamal. 19.7 m vs 17.2 m. On top = ahead.

Apply: Build, chart, compete

Everything today is in the sheet. Then Friday's contest.

1. Build the tip calculator exactly as shown, then add the shift log with 5 made-up shifts and a SUM.
2. Change the tip rate to 20%. Watch every row update. That's why the rate has its own cell.
3. Build the sprint sheet with Messi, Yamal, and Ronaldo-with-head-start. Chart all three.
4. Enter your jiu-jitsu data if you have it.
5. **Beat Dad Friday:** Week 2 is live. Both of you.
6. Check the cloud chip says **saved** (or save a progress file if you're offline).

Dad notes: Sheet Wizard badge: award it when he builds a working model from a blank sheet with no template, ideally the tip calculator without looking at the table above. If the sprint chart's lines don't start at the origin, check for a header row accidentally included in the data range.

Week 3 · Day 1: Two Lines Meet

Systems of equations, Achilles, the tortoise, and whether Yamal catches Ronaldo

Warm-up

Is the point **(2, 5)** on *both* lines $y = 2x + 1$ and $y = x + 3$?

- A. Yes, on both
- B. Only on the first
- C. Only on the second
- D. Neither

Answer: Both. $2(2) + 1 = 5$ and $2 + 3 = 5$. A point that works for two equations at once is called a solution of the system, and it's where the lines cross.

The Story

Two rules, one answer

DECODER

System of equations

Two (or more) equations that have to be true at the same time. The solution is the (x, y) pair that works for all of them. On a graph, it's where the lines cross.

$y = 2x + 1$ and $y = x + 3$ are both true at $(2, 5)$. Nowhere else. Two lines cross once (unless they're parallel, or the same line).

DECODER

Intersection

The point where two graphs cross. Same x , same y , both equations satisfied.

The moment two runners are at the same place at the same time.

WORKED EXAMPLE · THE CHASE

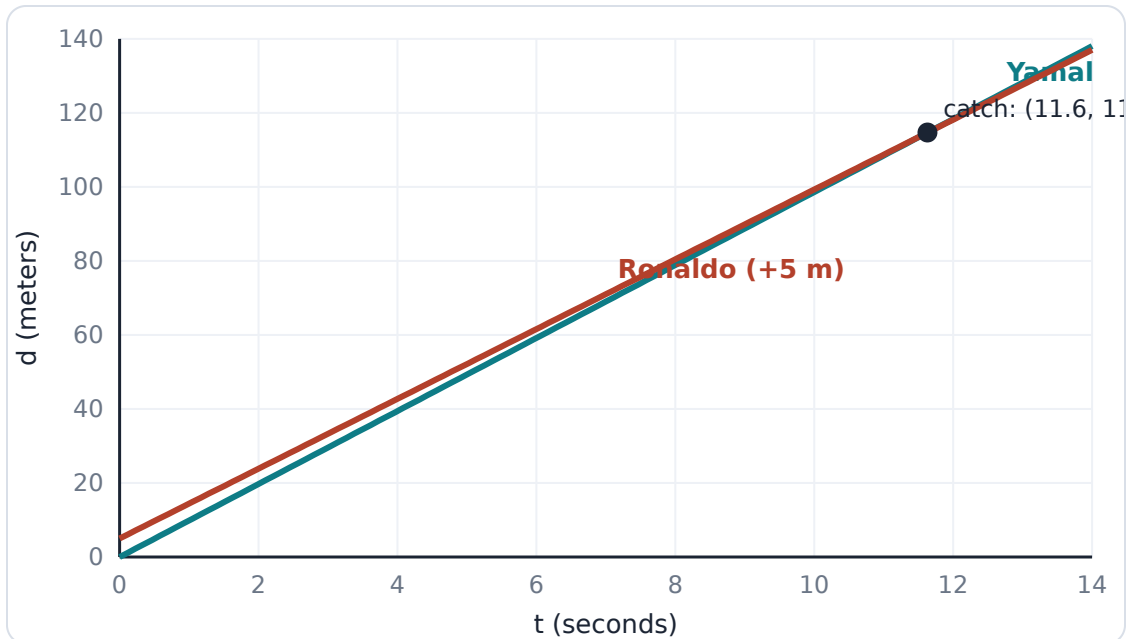
Ronaldo has a 5-meter head start and runs at his famous 9.43 m/s. Yamal starts behind him at 9.86 m/s. Both at top speed (it's a model). When does Yamal catch him?

Ronaldo: $d = 9.43t + 5$ Yamal: $d = 9.86t$

"Catch" means same distance at the same time: set them equal.

$9.86t = 9.43t + 5 \rightarrow 0.43t = 5 \rightarrow t \approx 11.6 \text{ s}$

Where? $d = 9.86 \times 11.6 \approx \mathbf{115 \text{ m}}$. A pitch is 105 m long. **Yamal doesn't catch him before the end line.** A 5-meter head start and a 0.43 m/s speed difference buys Ronaldo the whole field. That's why defenders play so close to the attacker: head starts are expensive.



Solving by graphing

Plot both lines, find the crossing, read off (x, y). Fast, visual, and only as accurate as your pencil. For "about 11 or 12 seconds" it's perfect. For "exactly when," you want algebra, which is tomorrow.

GREEK ORIGINS · ACHILLES AND THE TORTOISE

Zeno of Elea (around 450 BC) argued that Achilles, the fastest man alive, could never catch a tortoise with a head start. By the time Achilles reaches where the tortoise *was*, the tortoise has moved a bit. Achilles reaches *that* spot; the tortoise has moved again. Infinitely many steps, so (Zeno said) infinite time, so no catching.

A system of equations disagrees. Achilles at 10 m/s, tortoise at 0.1 m/s with a 100 m head start: $10t = 0.1t + 100$, so $t = 100 \div 9.9 \approx 10.1$ seconds. Done. The infinitely many steps add up to a finite time. Zeno's puzzle stayed unsolved for 2,000 years, until calculus explained why an infinite number of shrinking pieces can add up to something finite. That's Unit 5, and the Infinity badge. For now, Achilles wins, and so does algebra.

ON PAPER

Codex: *system of equations, intersection*. Graph the chase by hand on graph paper: t from 0 to 14, d from 0 to 140. Mark the crossing.

Practice

1. Achilles (10 m/s) vs. tortoise (0.1 m/s, 100 m head start). Solve $10t = 0.1t + 100$. Two decimals.

Answer: 10.1 s (± 0.05)

Hints: (1) Subtract $0.1t$ from both sides. (2) $9.9t = 100$.

Solution: $t = 100 \div 9.9 \approx \mathbf{10.10 \text{ s}}$. Zeno, refuted.

2. The chase: $9.86t = 9.43t + 5$. Solve for t (two decimals).

Answer: 11.63 s (± 0.1)

Hints: (1) Subtract $9.43t$: $0.43t = 5$.

Solution: $5 \div 0.43 \approx \mathbf{11.63 \text{ s}}$.

3. How far has Yamal run when he catches Ronaldo? (Use $t = 11.63$.) Nearest meter.

Answer: 115 m (± 1)

Hints: (1) $d = 9.86t$.

Solution: $9.86 \times 11.63 \approx \mathbf{115 \text{ m}}$.

4. The pitch is 105 m long. With a 5 m head start, does Yamal catch Ronaldo before the end line?

- A. Yes, at about 60 m
- B. No; the catch point is past the end line
- C. Yes, exactly at the line
- D. Can't tell from the model

Answer: No; the catch point is past the end line (B)

Hints: (1) Compare 115 m to 105 m.

Solution: No. $115 > 105$. Ronaldo reaches the end line first. (In reality Ronaldo is 41 and would be doing 33.95 km/h only in his memories, but the math is the math.)

5. $y = 2x + 1$ and $y = -x + 7$. The x-coordinate of the intersection?

Answer: 2

Hints: (1) Set them equal: $2x + 1 = -x + 7$. (2) $3x = 6$.

Solution: $x = \mathbf{2}$.

6. ...and the y-coordinate?

Answer: 5

Hints: (1) Plug $x = 2$ into either equation.

Solution: $2(2) + 1 = \mathbf{5}$. Check the other: $-2 + 7 = 5$. ✓ Intersection at (2, 5).

7. Two lines with the **same slope** and **different intercepts** (parallel). How many solutions does that system have?

- A. One
- B. Two
- C. None
- D. Infinitely many

Answer: None (C)

Hints: (1) Do parallel lines ever cross?

Solution: None. Two runners at exactly the same speed with a head start: the gap never changes. (Same slope *and* same intercept is the same line: infinitely many solutions.)

Apply: The head-start slider

Desmos turns the chase into a toy. Play with it until the answer to Friday's Labor is obvious.

1. Desmos: $y = 9.86x$ (Yamal) and $y = 9.43x + h$ (Ronaldo). Add a slider for h , range 0 to 20.
2. Drag h . Watch the crossing point slide along Yamal's line.
3. Add the end line: $y = 105$. It's horizontal.
4. Find the h where all three lines meet at one point: the head start that makes the catch happen exactly at the end line. Write it in your notebook. (Hint: it's between 4 and 5.)
5. Bonus: replace Ronaldo with Mbappé (10.44 m/s). What happens to the crossing? Why?

Dad notes: The answer to step 4 is about 4.6 m. Mbappé is faster than Yamal, so with any head start there's no crossing: the lines diverge. That's the parallel/diverging case in one drag of a slider.

Week 3 · Day 2: Substitution and Elimination

Two ways to solve a system exactly, and the Lego mix problem

Warm-up

If $y = 3x$ and $x + y = 12$, what is x ?

Answer: 3. Replace y with $3x$: $x + 3x = 12$, so $4x = 12$. You just did substitution.

The Story

Method 1: substitution

If one equation tells you what y is ($y = \text{something}$), replace y in the other equation with that something. Now there's only one variable, and you already know how to solve that.

WORKED EXAMPLE

$$y = 2x + 1 \text{ and } x + y = 10$$

Substitute: $x + (2x + 1) = 10 \rightarrow 3x + 1 = 10 \rightarrow 3x = 9 \rightarrow \mathbf{x = 3}$. Then $y = 2(3) + 1 = \mathbf{7}$.
Check the second equation: $3 + 7 = 10$. ✓

Method 2: elimination

Add (or subtract) the two equations so one variable disappears. Works when the variables are lined up and one of them has matching (or opposite) amounts.

WORKED EXAMPLE

$$\begin{array}{l} 2x + y = 11 \\ x - y = 1 \end{array}$$

Add them straight down. The $+y$ and $-y$ cancel: $3x = 12 \rightarrow \mathbf{x = 4}$. Then $4 - y = 1 \rightarrow \mathbf{y = 3}$. Check: $2(4) + 3 = 11$. ✓

If nothing cancels, multiply one equation first so something does. Both methods give the same answer; pick whichever is less typing.

WORKED EXAMPLE · THE LEGO MIX

A bag holds 100 pieces: 2×4 bricks at 10 cents and 1×1 plates at 3 cents. The bag cost \$7.20. How many of each?

$$b + p = 100 \quad 0.10b + 0.03p = 7.20$$

Substitution: $p = 100 - b$. Then $0.10b + 0.03(100 - b) = 7.20 \rightarrow 0.10b + 3 - 0.03b = 7.20 \rightarrow 0.07b = 4.20 \rightarrow \mathbf{b = 60 \text{ bricks, } p = 40 \text{ plates}}$. Check: $6.00 + 1.20 = 7.20$. ✓

This is the shape of half the problems in business: two unknowns, a count and a cost.

MAT MATH · SCORING

IBJJF scoring: takedown 2 points, mount 4 points (among others). Hudson scores 5 times for 14 points, all takedowns and mounts. How many of each? $t + m = 5$ and $2t + 4m = 14$. Elimination: double the first ($2t + 2m = 10$) and subtract from the second: $2m = 4 \rightarrow \mathbf{m = 2}$, so $\mathbf{t = 3}$. Three takedowns, two mounts. A good day.

ON PAPER

Codex: *substitution, elimination*. Solve every practice problem below on paper with the method named, and write the check.

Practice

1. $y = 2x + 1$ and $x + y = 10$. Find x .

Answer: 3

Hints: (1) Substitute $2x + 1$ for y in the second equation. (2) $3x + 1 = 10$.

Solution: $x = 3$.

2. ...and y .

Answer: 7

Hints: (1) $y = 2(3) + 1$.

Solution: $y = 7$. Check: $3 + 7 = 10$. ✓

3. $2x + y = 11$ and $x - y = 1$. Find x by elimination.

Answer: 4

Hints: (1) Add the equations; the y 's cancel. (2) $3x = 12$.

Solution: $x = 4$ (and $y = 3$).

4. Lego mix: 100 pieces, bricks at 10¢ and plates at 3¢, total \$7.20. How many **bricks**?

Answer: 60 bricks

Hints: (1) $b + p = 100$, so $p = 100 - b$. (2) $0.10b + 0.03(100 - b) = 7.20$. (3) $0.07b = 4.20$.

Solution: **60 bricks**, 40 plates.

5. Fortnite: you have **500 building materials** total, wood and brick. You have **100 more wood than brick**. How much **brick**?

Answer: 200 brick

Hints: (1) $w + b = 500$ and $w = b + 100$. (2) Substitute: $(b + 100) + b = 500$.

Solution: $2b + 100 = 500 \rightarrow b = 200$, $w = 300$.

6. Scoring: takedowns 2 points, mounts 4 points. **5 scores, 14 points**. How many **mounts**?

Answer: 2 mounts

Hints: (1) $t + m = 5$ and $2t + 4m = 14$. (2) Double the first equation and subtract.

Solution: **2 mounts**, 3 takedowns.

7. Camp Half-Blood store: 3 ambrosia + 2 nectar = 13 drachmas. 1 ambrosia + 2 nectar = 7 drachmas. Price of **one ambrosia**?

Answer: 3 drachmas

Hints: (1) The nectar amounts match. Subtract the second equation from the first. (2) $2a = 6$.

Solution: **3 drachmas** per ambrosia; nectar is 2. Don't eat too much; it's fatal to mortals, which is a strong pricing strategy.

Apply: The checker

A Sheet that checks systems for you is a good way to make sure you actually understand what "solution" means.

1. New tab "System checker." B1 = your x, B2 = your y.
2. B4: the left side of equation 1 as a formula using \$B\$1 and \$B\$2 (e.g. $=2*B1+B2$). C4: what it should equal (11). D4: $=B4=C4$. It shows TRUE or FALSE.
3. Row 5: the same for equation 2.
4. Type in your answers from the practice problems. Both rows should say TRUE. If one says FALSE, you've found a mistake before I did.
5. Claude: ask it for a hint (not the answer) on the Camp Half-Blood problem, as if you hadn't solved it. Did the hint match the method you used?

Dad notes: The TRUE/FALSE trick generalizes: it's how programmers write tests. If he's into it, mention that.

Week 3 · Day 3: The Whole Language

Review, the Feynman technique, and the mistakes everyone makes

Warm-up

Slope of $y = -3x + 8$?

Answer: -3. Falling 3 for every 1 across.

The Story

The map of Unit 1, on one page

- 1 A function is a machine.** Input $\rightarrow$ rule $\rightarrow$ output. $f(x)$ is "f of x," not f times x.
- 2 A graph is its picture.** Every (input, output) is a point. Straight line = steady rate.
- 3 Slope is the rate.** Rise over run. On a position graph, slope is velocity.
- 4 $y = mx + b$.** b is where it starts. m is how fast it changes.
- 5 Solving is undoing.** Same move on both sides. Undo the last thing first. Check by plugging in.
- 6 Two machines meet** where their outputs match. Set them equal, or substitute, or eliminate.

The mistakes gallery

- Treating $f(x)$ as multiplication.
- Slope upside down (run over rise). Remember: rise is y, and y is the vertical one.
- Losing the minus sign on a negative slope.
- Forgetting to divide km/h by 3.6.
- Writing 18 instead of 0.18 for 18%.
- Not checking. The check takes five seconds and would have caught all of the above.

THE STORY · THE FEYNMAN TECHNIQUE

Feynman had a habit that made him the best explainer in physics: whenever he learned something, he'd try to explain it in plain words, as if to a first-year student. Wherever he got stuck or had to use a fancy term he couldn't unpack, that was the spot he didn't actually understand yet. Then he'd go back and fix that spot. Four steps:

- 1** Pick the idea. (Today: **slope**.)
- 2** Explain it out loud to someone who doesn't know it, in plain words, with a drawing. No notes.

3 Notice where you stumble or reach for jargon. That's the gap.

4 Go back, fill the gap, explain again.

The Feynman badge is exactly this: explain one idea from the unit to Dad, no notes, clearly enough that he could explain it to someone else. It's the hardest badge in the program and the one that means the most.

MATHEMAGIC · SHOW PREP

Two tricks for Friday's family show: your own number trick (proved), and the **Oracle of the Crossing**: give Dad two lines (two runners, two prices, whatever) and announce where they'll meet *before* he can graph it. You already solved it on paper. He'll think it's magic. It's substitution.

ON PAPER

Codex check: you should have at least 12 Unit 1 terms with your own definitions. Fill in any you skipped. Then do the review set below without hints if you can.

Practice

1. $f(x) = -2x + 9$. Find $f(3)$.

Answer: 3

Hints: (1) $-2(3) + 9$.

Solution: $-6 + 9 = 3$.

2. Slope through **(2, 3)** and **(6, 15)**.

Answer: 3

Hints: (1) Rise 12, run 4.

Solution: $12 \div 4 = 3$.

3. Slope 3, passes through $(0, -4)$. Equation?

A. $y = -4x + 3$

B. $y = 3x - 4$

C. $y = 3x + 4$

D. $y = -3x - 4$

Answer: $y = 3x - 4$ (B)

Hints: (1) Intercept is -4 .

Solution: $y = 3x - 4$.

4. $6x - 9 = 3x + 12$

Answer: 7

Hints: (1) $3x - 9 = 12$. (2) $3x = 21$.

Solution: $x = 7$. Check: $33 = 33$. ✓

5. Mbappé, **37.6 km/h**, fastest at the 2026 World Cup. In m/s, two decimals:

Answer: 10.44 m/s (± 0.02)

Hints: (1) $\div 3.6$.

Solution: **10.44 m/s**. About one meter per second faster than Yamal, which over a 30 m sprint is roughly 3 meters.

6. Storm phase 4: 7 HP/s. From 100 HP, how many seconds until zero? One decimal.

Answer: 14.3 s (± 0.1)

Hints: (1) $100 - 7t = 0$.

Solution: $100 \div 7 \approx$ **14.3 s**.

7. $y = x + 4$ and $y = 3x$. The x at the intersection?

Answer: 2

Hints: (1) $x + 4 = 3x$.

Solution: $4 = 2x \rightarrow x = 2$ ($y = 6$).

8. 1,000 Knuts is how many **whole Galleons**? (493 Knuts per Galleon.)

Answer: 2 Galleons

Hints: (1) $1000 \div 493$.

Solution: 2.03, so **2 Galleons** with 14 Knuts left over. The goblins keep the 14.

9. Minecraft sprinting (**5.612 m/s**): time for 500 blocks, one decimal.

Answer: 89.1 s (± 0.2)

Hints: (1) $500 \div 5.612$.

Solution: **89.1 s**.

10. A shot covers **1.8 m in 0.6 s**. Speed?

Answer: 3 m/s (± 0.01)

Hints: (1) Distance $\div$ time.

Solution: **3 m/s**. If your real data from class beat this, you're faster than the textbook double-leg.

Apply: Teach it back

The Feynman badge attempt. Two minutes, one idea, one drawing, no notes.

1. Choose: slope, or intercept, or "why two lines crossing means the runners meet."
2. Explain it to Dad. He plays someone who has never heard of it. He's allowed to ask "why?" three times.
3. Notice where you stumbled. Write that spot in the notebook.
4. Fix it (re-read the story, ask Claude to explain that one spot) and explain again tomorrow.
5. Finish the Codex: 12 Unit 1 terms in your own words. Each earns 10 XP on the Codex page.

Dad notes: Feynman badge standard: could *you* now explain it to someone else using only what he said? If yes, award it. If he says "it's just the number in front of x ," ask "but what does it *mean*?" and wait. The pause is the lesson.

Feynman Badge prompt: Explain **slope** to Dad as if he's never seen a graph. Use a drawing. Use the words "for every." No notes. If he can repeat it to someone else afterward, the badge is yours.

Week 3 · Day 4 · Lab Day: Lab Day: The Catch

Labor 1, Quest 2, the family show, and the Cartographer badge

Warm-up

15% of 60?

Answer: 9. 10% is 6, half of that is 3, together 9.

The Story

Today's order of business

- 1 Labor 1: The Catch.** Five parts, no hints. Algebra on paper first, then the Sheets proof. Beat it: Cartographer badge.
- 2 Quest 2.** Four clues. Dad has hidden them. The lock wants three digits.
- 3 The family show.** Your number trick, then the Oracle of the Crossing. Willman badge if it's new and explained.
- 4 Beat Dad, Week 3.**
- 5 Check the cloud chip says saved.**

THE STORY · WHAT YOU CAN DO NOW THAT YOU COULDN'T THREE WEEKS AGO

Read a graph. Compute a rate. Turn a sentence into an equation and an equation into an answer. Predict where two moving things meet. Build a spreadsheet model that updates itself. Design a magic trick from a formula. Explain slope to a human being.

That's most of Algebra I. It took three weeks because you already spoke the language; you just hadn't been told what the words meant. Unit 2 is money, and it's where the lines start to bend.

Practice

1. Warm the engine for the Labor: convert **35.5 km/h** (Yamal) to m/s, two decimals.

Answer: 9.86 m/s (± 0.02)

Hints: $(1) \div 3.6$.

Solution: 9.86 m/s.

2. Yamal's speed minus Ronaldo's ($9.86 - 9.43$): the **closing speed**, in m/s.

Answer: 0.43 m/s (± 0.01)

Hints: (1) Subtract.

Solution: **0.43 m/s.** That's how fast the gap shrinks. Every catch-up problem is "gap $\div$ closing speed."

3. Using gap $\div$ closing speed: with a **10 m** gap, how long to close it? One decimal.

Answer: 23.3 s (± 0.1)

Hints: (1) $10 \div 0.43$.

Solution: **23.3 s.** Longer than any sprint on a pitch. Big head starts are basically permanent at these speeds.

Apply: Labor, Quest, Show

Go to the Labor page (Unit 1 $\rightarrow$ Labor 1). Then the Quest page. Then the living room.

1. Labor 1: The Catch. Paper first for every part.
2. Sheets proof: a table with t from 0 to 15, Yamal and Ronaldo columns, a chart, and the row where Yamal pulls ahead highlighted.
3. Quest 2: clue 1 is on the Quest page.
4. Family show: two tricks, both explained after.
5. Beat Dad Week 3, then check the cloud chip says **saved**.

Dad notes: Quest 2 setup is on the Quest page in Dad mode. Award Willman for the show if the Oracle trick is explained (it's substitution: he solved the crossing before you graphed it). This is a good night to print his Unit 1 Codex page and put it on the fridge.

Labor 1: The Catch

Cristiano Ronaldo (9.43 m/s, his 2018 World Cup number) has a **6-meter head start** on Lamine Yamal (35.5 km/h at the 2026 World Cup). Both at top speed, straight line, full pitch. Does Yamal catch him? Prove it three ways: algebra here, a graph on paper, a table in Sheets. No hints.

Part 1. Part 1. Convert Yamal's 35.5 km/h to m/s (two decimals).

Answer: 9.86 m/s (± 0.02)

Solution: $35.5 \div 3.6 = 9.86$ m/s.

Part 2. Part 2. Write both distance equations (Ronaldo: $d = 9.43t + 6$; Yamal: $d = 9.86t$), set them equal, and solve for the catch time t . Two decimals.

Answer: 13.95 s (± 0.15)

Solution: $9.86t = 9.43t + 6 \rightarrow 0.43t = 6 \rightarrow t = 13.95$ s.

Part 3. Part 3. How far from Yamal's starting point is the catch? Nearest meter.

Answer: 138 m (± 2)

Solution: $9.86 \times 13.95 \approx 137.6$ m.

Part 4. Part 4. The pitch is 105 m long. Verdict?

- A. Yamal catches him around midfield
- B. Yamal catches him just before the end line
- C. Yamal does not catch him; Ronaldo reaches the end line first
- D. They arrive together

Answer: Yamal does not catch him; Ronaldo reaches the end line first (C)

Solution: 138 m > 105 m. Ronaldo starts 6 m in, so he has 99 m to the end line: $99 \div 9.43 = 10.5$ s. At that moment Yamal has run $9.86 \times 10.5 = 103.5$ m, still behind. No catch.

Part 5. Part 5. What head start (in meters, one decimal) would make Yamal catch Ronaldo *exactly* at the 105 m end line? (Hint you're allowed: how long does Yamal take to run 105 m? How far does Ronaldo get in that time?)

Answer: 4.6 m (± 0.15)

Solution: Yamal runs 105 m in $105 \div 9.86 = 10.65$ s. In 10.65 s Ronaldo covers $9.43 \times 10.65 = 100.4$ m. So the head start must be $105 - 100.4 \approx 4.6$ m. Any bigger and Ronaldo wins.

Quest 2: The Cartographer's Quest (Dad setup)

Four clues. Every answer is a number that points somewhere in the house. The lock wants the last digit of the first three answers.

Hide in order. A 3-digit luggage lock reads **2-6-5** (last digits of 52, 26, 365). A dial padlock reads **26-5-4** (clue 2's answer, the last digit of clue 3, clue 4's answer); if you use a dial, tell him that rule. The box goes where clue 4 points.

#	Clue	Answer	Hiding spot for the next clue
1	Solve $5x - 8 = 252$. A full deck has exactly this many. The next clue is in the deck.	52	Inside his deck of playing cards (the magic one).
2	$y = 3x + 2$. Find y when $x = 8$. The alphabet has this many letters, and they're all in one heavy book on the shelf.	26	In the dictionary (or the biggest book on the shelf).
3	$x + y = 730$ and $y = x$. Find x . Every one of those is a square on the wall. Look at this month.	365	Behind or on the calendar.
4	$2x + y = 12$ and $x - y = 0$. Find x . The car has this many. So does the box's hiding place. The lock wants the last digit of your first three answers.	4	The box goes in the garage, by the car (or by a wheel).

Lock: Luggage lock: **2-6-5**. Dial padlock: **26-5-4** (tell him the rule: clue 2, last digit of clue 3, clue 4).